EXPERIMENT - 7

24/9/26 SLIDING WINDOW

AIM:

To write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL and to simulate the flow of frames from one node to another.

CODE:

Sender.py:

```
import time
window_size = int(input("Enter window size: "))
message = input("Enter the message")
frames = []
for i in range(0, len(message), 5):
    data = message[i:i+5]
    frames.append([i//5, data])
print("\n Frames created: ")
for frame in frames:
    print("Frame No: ", frame[0], "Data: ", frame[1])
start = 0
while start < len(frames):
    end = min(start + window_size, len(frames))
    print("\n Sending frames: ")
    with open("sender_buffer.txt", "w") as file:
        for i in range(start, end):
            file.write(f"{frames[i][0]} {frames[i][1]}")
            print("Sending frame", frames[i][0], frames[i][1])
    print("\n Waiting for Ack")
    time.sleep(5)
    try:
        with open("receiver_buffer.txt", "r") as file:
            ack = int(file.read().strip())
        print("Ack received:", ack)
    except:
        print("NO ACK received")
        continue
    if ack == end - 1:
        print("Ack is correct")
        start = end
    else:
        print("NACK! Wrong Ack!")
        print("Resending frames from frame", ack + 1)
        if ack >= start:
            start = ack + 1
        else:
            start = start
print("All frames transmitted Sequency")
```

Receiver.py:

```
import time
import os
print("Receiver started:")
expected_frame = 0
old_buffer = ""
while True:
    if not os.path.exists("sender_buffer.txt"):
        time.sleep(1)
        continue
    with open("sender_buffer.txt", "r") as file:
        data = file.read()
    if data == old_buffer:
        time.sleep(1)
```

```
old_buffer = data.
lines = data.strip().split("\n").
print("\n Received Frames").
last_ack = expected_frame - 1
for line in lines:
    parts = line.split(" ", 1).
    frame_no = int(parts[0]).
    frame_data = parts[1]
    print("Frame ", frame_no, ": ", frame_
    if frame_no == expected_frame:
        print("Frame accepted").
        last_ack = frame_no.
        expected_frame += 1.
    else:
        print("Wrong frame!").
        break
print("Sending Ack: ", last_ack)
with open("receiver_buffer.txt", "w") as file
    file.write(str(last_ack))
last_frame = int(lines[-1].split(" ", 1)[0]).
if last_ack == last_frame:
    print("\n All frames received successfu
    break
time.sleep(1).
```

RESULT: The sliding window Protocol was successfully implemented using Python. Frames were transmitted and acknowledged successfully, demonstrating reliable data transm